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(1) Research Program/Project/Study Title: <i>Physio-chemical Properties of CAPSU Formulated Soil Ameliorants</i>	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☐ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☒ Natural/BiologicalSciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster	(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☒ CSRDC Crop Science Research Development Center
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(4) Site of implementation	
Municipality/City:	Mambusao
Province:	Capiz
Region:	VI

(5) Classification Research: ☐ Basic ☒ Applied Development: ☐ Pilot Testing ☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☒ Single Authorship/Proponent ☐ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)
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(7) Sector/Commodity Soil Ameliorant	(8) Discipline/Field of Specialization Agriculture
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(9A) Abstract (This has a maximum of 250 words. It contains 1-2 sentences of rationale, followed by objectives, brief methodology, major findings, and implications of impact of the study.)

The experiment was conducted at the Organic Agriculture Project of Capiz State University - Burias Campus, located in Burias, Mambusao, Capiz. The study tested variants of CAPSU Formulated Soil Ameliorants (CAFSA) on cow manure, hog manure, and chicken dung. Data were presented in line graphs to determine trends.

The results revealed that the highest temperature of 59.66°C was recorded in CAFSA with Chicken Dung Variant 2 (CAFSA V2) on Day 16, while the lowest temperature (26°C) was in CAFSA with Hog Manure Variant 2 and 3 (CAFSA V2 and V3) at on Day 31. Initial stability was observed on the 8th day of mixing for all CAFSA formulations and variants. A temperature decline was noted in all the CAFSA formulations and variants starting on Days 4-5. With regards to moisture content, CAFSA with hog manure variant 1 (CAFSA V1) showed a significant moisture drop to 4.44 on Day 29, while Chicken Dung manifested the most consistent moisture content across all variants. Likewise, all CAFSA formulations and variants, showed an initial increase in pH, with CAFSA containing chicken dung variant 2 (CAFSA V2) reaching the highest pH of 6.78. CAFSA with hog manure had the lowest initial pH showing steady increase across variants. In terms pH and organic matter (OM) CAFSA with hog manure variant 2 showed the highest reading. While highest phosphorus was recorded from CAFSA with hog manure variant 1, and highest potassium in variant 2. For CAFSA with cow manure the highest pH was recorded in variant 2 with decreasing OM, increasing phosphorus across variants, and highest potassium in variant 2. CAFSA with chicken dung had the highest pH in variant 2, highest OM in V3, stable nitrogen across versions, highest phosphorus in variant 3, and highest potassium in variant 2.

(9B) Keywords (Write 3-6 words you used in the study, preferably not found in the title because words in the title are already given keywords. This would also include process, or methodology employed.)

organic farming, organic, soil ameliorants, CAFSA, physiochemical properties

(10) Introduction (This includes the rationale, background of the study, and objectives which are not divided by subheadings. Incorporate at least three (3) reviews)

The integration of soil ameliorants into agricultural practices has been widely recognized for its capacity to enhance soil quality and boost crop productivity. The Capiz State University (CAPSU) has formulated a unique blend of soil ameliorants comprising animal manure, garden soil, compost, vermicast, rice bran, carbonized rice hull, effective microorganism inoculant, fermented plant juice, fish amino acid, indigenous microorganisms, lactic acid, molasses, and water.

Soil ameliorants, by definition, are materials added to soil to improve its physical, chemical, and biological properties (Zhang et al., 2013). The CAPSU mix is strategically designed to address various soil deficiencies and enhance soil structure, nutrient content, and microbial activity.

Animal manure is a rich source of organic matter and essential nutrients such as nitrogen, phosphorus, and potassium. It plays a significant role in improving soil fertility, water-holding capacity, and microbial activity (Goss et al., 2013). Studies indicate that the addition of animal manure enhances soil organic carbon content and improves soil physical properties, resulting in better plant growth and yield (Chaudhry et al., 2009; Nash et al., 2012).

Garden soil contributes to the mineral content and texture of the soil, supporting root development and nutrient uptake (Brady & Weil, 2002). It provides a stable base that helps

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maintain soil structure and serves as a medium for the other components of the soil ameliorant mix.

Compost is produced from decomposed organic matter and is rich in humus, which enhances nutrient availability and promotes beneficial microbial populations (Bernal et al., 2009). The application of compost improves soil structure, increases nutrient content, and enhances soil moisture retention, leading to improved crop performance (Meena et al., 2016).

Vermicast, or worm castings, is highly regarded for its nutrient content and ability to enhance soil structure and microbial activity (Arancon et al., 2005). Research shows that vermicast application improves soil fertility and plant growth by increasing nutrient availability and promoting beneficial soil microorganisms (Lazcano & Domínguez, 2011).

Rice bran, a by-product of rice milling, is rich in nutrients and organic matter. It improves soil fertility and supports plant growth (Yamamoto et al., 2014). Carbonized rice hull enhances soil aeration and water retention, providing a stable source of organic carbon that supports microbial activity (Sombroek et al., 1993).

EM inoculants introduce beneficial microbes into the soil, enhancing nutrient cycling, suppressing harmful pathogens, and promoting plant health (Higa & Parr, 1994). The use of EM in soil management has been associated with improved soil fertility and increased crop yields (Kopittke et al., 2019).

These components provide organic nutrients and growth-promoting compounds, enhancing plant vigor and yield. Fermented plant juice is rich in plant hormones and enzymes, while fish amino acid supplies amino acids and micronutrients essential for plant growth (Kahrl et al., 2010).

Indigenous microorganisms promote a healthy soil microbial ecosystem, crucial for nutrient cycling and soil fertility. Lactic acid helps in the fermentation process, enhancing the availability of nutrients and promoting beneficial microbial activity (Tamang et al., 2016).

Molasses serves as a carbon source, supporting microbial activity and nutrient availability in the soil. It enhances the decomposition of organic matter and promotes the growth of beneficial soil microorganisms (Marschner, 2012).

Water is essential for the activation and distribution of the soil ameliorants in the soil. Proper irrigation ensures that the ameliorants are effectively integrated into the soil, providing consistent moisture and nutrient availability for optimal plant growth (Fereses & Soriano, 2007).

The synergistic effect of these components in the CAPSU formulated soil ameliorants leads to significant improvements in soil physicochemical properties. Enhanced soil structure, increased nutrient availability, and improved microbial activity create an optimal environment for plant growth, resulting in higher yields and better-quality produce (Marschner, 2012).

These components work together to improve soil fertility, structure, and biological activity, leading to improved soil health and crop productivity. Continued research and application of such tailored soil ameliorants can contribute to sustainable agricultural practices and enhanced crop yields.

Objectives of the Study

The study was conducted to determine the physicochemical properties of CAPSU Formulated Soil Ameliorants (CAFSA). Specifically, the study aims to:

1. Determine the temperature, moisture, and pH level of CAPSU Formulated Soil Ameliorants (CAFSA); and
2. Evaluate which variant of CAPSU Formulated Soil Ameliorants (CAFSA) exhibits the

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best physic-chemical properties.

(11) Materials and Methods or Methodology (This includes the materials, methods, duration & location of the study. Materials and Methods are headings used in Natural Science researches and methodology is Social Sciences. Presentation of materials may not necessarily be in separate heading but may be incorporated in the specific methods used. Justify paragraphs with main headings in caps.)

The experiment was conducted at the Organic Agriculture Project of Capiz State University - Burias Campus, located in Burias, Mambusao, Capiz. The treatments tested were Variants 1, 2, and 3 of CAPSU Formulated Soil Ameliorants (CAFSA) containing cow manure, hog manure, and chicken dung. The data were presented in line graphs to determine the trends.

(12) Results and Discussions (This part contains presentation of results (data) with reference to the table and figure, to be followed by discussion, with support of reviewed literature. Sequence shall be based on the objectives of the study. Tables are formatted without the vertical lines. Horizontal lines are set at the top and bottom portions as well as below table headings and above major subdivisions. Justify paragraphs with main headings in caps.)

Temperature of CAFSA Formulations for Variant 1 (CAFSA V1)

The data revealed that the daily temperature levels of CAFSA Variant 1 of cow manure, hog manure, and chicken dung for 31 days were as follows: CAFSA with cow manure started at a temperature of 28.67°C, peaked at 53.55°C on day 8, and then gradually declined to 27°C by day 31. CAFSA with hog manure started at 27.33°C, reached the highest of 53.33°C on Day 10, and then steadily decreased to 26.33°C until day 31. Chicken Dung started at 29.56°C, with temperatures peaking at 56.22°C on day 7, and ended at 31.66°C on day 31. An initial increase in temperature was observed for all variants, peaking between days 7 and 10, followed by a gradual decline towards the end of the period. The highest temperature recorded was 56.22°C for CAFSA with chicken dung on day 7, while the lowest was 26.33°C for CAFSA with hog manure on day 31.

Temperature CAFSA Formulations for Variant 2 (CAFSA V2)

The data revealed that the daily temperature levels of CAFSA V2 for cow manure, hog manure, and chicken dung up to 31 days were as follows: CAFSA with cow manure started at a temperature of 28.56°C, peaked at 52.66°C on day 14, and then gradually declined to 26.33°C by day 31. CAFSA with hog manure started at 27.44°C, reached as high as 53.33°C on Day 4, and then steadily decreased to 26°C by day 31. CAFSA with chicken dung started at 29.11°C on day 1, with temperatures peaking at 59.66°C on day 16, and ended at 30.44°C on day 31. An initial increase in temperature was observed for all formulations, peaking between days 4 and 16, followed by a gradual decline towards the end of the period. The highest temperature recorded was 59.66°C for CAFSA with chicken dung on day 16, while the lowest was 26°C for CAFSA with hog manure on day 31.

Temperature CAFSA Formulations for Variant 3 (CAFSA V3)

The data showed that the daily temperature levels of CAFSA V3 for cow manure, hog manure, and chicken dung for 31 days were as follows: CAFSA with cow manure started at a temperature of 28.22°C, peaked at 55°C on day 11, and then gradually declined to 26.11°C by day 31. CAFSA with hog manure started at 27.44°C, reached as high of 52.55°C on day 5, and then steadily decreased to 26°C by day 31. CAFSA with chicken dung started at 28.89°C on day 1, with temperatures peaking at 59.33°C on day 14, and ended at 31.88°C on day 31. An initial increase in temperature was observed for CAFSA formulations, peaking between days 5 and 14, followed by a gradual decline towards the end of the period. The highest temperature recorded was 59.33°C for CAFSA with chicken Dung on day 14, while the lowest was 26°C for CAFSA with hog manure on days 25 and 31.

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Moisture Content of CAFSA Formulations for Variant 1 (CAFSA V1)

The data showed that the daily moisture levels of CAFSA V1 for cow manure, hog manure, and chicken dung up to 31 days were as follows: CAFSA with cow manure started at a moisture level of 8, gradually declined to as low as 5.13 on day 24 before slightly increasing towards the end. CAFSA with hog manure also started at 8, showed a steady decrease with significant drops at day 21, and reaching a low of 4.44 on day 29. CAFSA with chicken dung maintained a relatively stable moisture level of 8 for most days, with minor fluctuations but returning again at 8 towards the end of the observation period. An initial stability around 8 was observed for all CAFSA formulations, with decline starting around day 4, and variable moisture and eventually significant drop was shown by CAFSA with hog manure. CAFSA with chicken dung has the most consistent moisture levels, maintaining a reading of 8 for 22 out of the 31 days.

Moisture Content of CAFSA Formulations for Variant 2 (CAFSA V2)

The data revealed that the daily moisture levels of CAFSA V2 for cow manure, hog manure, and chicken dung for 31 days were as follows: CAFSA with cow manure started at a moisture level of 8, and gradually declined to a low of 5.13 on day 24. CAFSA with hog manure also started at 8, showed a steady decrease with significant drops after day 21, reaching as low of 4.44 on day 29. CAFSA with chicken dung maintained a relatively stable moisture level of 8 for most days, with minor fluctuations on the 15th day but stabilizing again at 8 towards the end of the month. For CAFSA with hog manure, a moisture content of 8 was observed, with a decline on day 5 up to the middle of the observation period. CAFSA with chicken dung had the most consistent moisture levels.

Moisture Content of CAFSA Formulations for Variant 3 (CAFSA V3)

The data outlined the daily moisture levels of CAFSA V3 for cow manure, hog manure, and chicken dung for 31 days. CAFSA with cow manure started at a moisture level of 8, gradually declined to as low as 6.44 on day 31. CAFSA with hog manure also started at 8, showed a steady decrease with significant drops after day 22, reaching a low of 5.44 on days 30 and 31. CAFSA with chicken dung maintained a relatively stable moisture level of 8 for most days, with minor fluctuations middle of the observation period but stabilizing again at 8 towards the end of the 31 days. Initial moisture of 8 was observed for all CAFSA formulations, with a decline starting the day 4 up to the middle period. CAFSA with hog manure showed slight decline in the moisture content towards the end of the observation period and a significant drop on day 31. CAFSA with chicken Dung had the most consistent moisture levels, maintaining a level of 8 for 22 out of the 31 days.

pH Level CAFSA Formulations for Variant 1 (CAFSA V1)

The pH levels for CAFSA V1 for cow manure, hog manure, and chicken dung were measured over 31 days. For CAFSA with cow manure, pH values ranged from 5.06 to 6.13, with a general upward trend and some fluctuations throughout the period. CAFSA with hog manure showed pH levels between 4.92 and 6.36, demonstrating a steady increase over time. CAFSA with chicken dung had pH levels varying from 4.2 to 6.44, with notable fluctuations and a general upward trend towards the end. All CAFSA formulations experienced an initial increase in pH, peaking around the middle of the observation period, and followed by a gradual rise towards the end.

pH Level CAFSA Formulations for Variant 2 (CAFSA V2)

The pH levels for CAFSA V2 containing cow manure, hog manure, and chicken dung were recorded over a 31-day period. For CAFSA with cow manure, the pH values ranged from 5.04 to 6.01, showing a general upward trend with some fluctuations. CAFSA with hog manure had pH levels between 5.2 and 6.28, displaying a steady increase throughout the measurement period. CAFSA with chicken dung exhibited pH values from 4.25 to 6.78, with notable fluctuations and a general upward trend as the period progressed. All CAFSA formulations experienced an initial decrease in pH followed by a gradual increase. The peak pH levels for CAFSA with chicken dung were higher compared to

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CAFSA with cow manure and hog manure. This trend suggested that while there was a general rise in pH levels over time, each CAFSA formulations Variant 2 responded differently.

pH Level CAFSA Formulations for Variant 3 (CAFSA V3)

The pH levels for CAFSA V3 with cow manure, hog manure, and chicken dung were monitored over a 31-day period. For CAFSA with cow manure, the pH values ranged from 5.06 to 6.17, with a general upward trend observed, particularly in the latter half of the observation period. CAFSA with hog manure showed pH levels between 5.01 and 6.17, with a steady increase over time. CAFSA with chicken dung exhibited pH values from 4.46 to 6.37, demonstrating significant fluctuations but an overall upward trend. An initial decrease in pH was observed for CAFSA Formulations, followed by a rise that became more pronounced towards the end of the observation period. CAFSA with chicken dung showed the highest pH values and the most variability. The data indicated that while pH levels generally increased over time, each CAFSA formulations exhibited unique trends, with chicken dung showing the most fluctuation.

pH, Organic Matter (OM) , Nitrogen, Phosphorus and Potassium (NPK) Content of CAFSA Formulations Variant 1 (CAFSA V1)

The organic matter (OM), nitrogen (N), phosphorus (P), and potassium (K) levels of CAFSA V1 having cow manure, hog manure, and chicken dung were analyzed. The pH levels were 5.82 for CAFSA with hog manure, 6.57 for cow manure, and 6.92 for chicken dung, indicating a general increase for hog manure to chicken dung.

Organic matter content was lowest in CAFSA with hog manure at 2.21%, slightly higher in CAFSA with cow manure at 2.85%, and highest in CAFSA with chicken dung at 2.88%. Nitrogen content was lowest in CAFSA with hog manure at 0.72%, slightly higher in CAFSA with cow manure at 0.80%, and highest in CAFSA with chicken dung at 1.06%. Phosphorus and potassium contents showed similar trends: CAFSA with hog manure had 0.78% phosphorus and 1.09% potassium, CAFSA with cow manure had 0.82% phosphorus and 1.24% potassium, and CAFSA with chicken dung had 1.04% phosphorus and 1.32% potassium. The results highlighted that CAFSA with chicken dung generally had the highest pH, OM, N, P, and K levels, followed by CAFSA with cow manure and hog manure.

pH, Organic Matter (OM) , Nitrogen, Phosphorus and Potassium (NPK) Content of CAFSA Formulations Variant 2 (CAFSA V2)

The pH, organic matter (OM), nitrogen (N), phosphorus (P), and potassium (K) contents of CAFSA V2 containing cow manure, hog manure, and chicken dung were evaluated. CAFSA with cow manure exhibited a pH of 6.63, with 2.85% OM, 0.82% N, 0.86% P, and 1.21% K. CAFSA with hog manure had a pH of 6.28, with 2.32% OM, 0.71% N, 0.79% P, and 1.10% K. CAFSA with Chicken Dung had the highest values with a pH of 6.72, 3.08% OM, 1.09% N, 1.08% P, and 1.40% K. The data indicated that Chicken Dung generally had the highest pH, OM, N, P, and K contents, followed by Cow Manure and Hog Manure. Chicken Dung showed significant advantages in nutrient content compared to the other two manures.

pH, Organic Matter (OM) , Nitrogen, Phosphorus and Potassium (NPK) Content of CAFSA Formulations Variant 3 (CAFSA V3)

The pH, organic matter (OM), nitrogen (N), phosphorus (P), and potassium (K) levels of CAFSA of different formulations for V3 were analyzed. CAFSA with cow manure had a pH of 6.12, with 2.82% OM, 0.81% N, 0.80% P, and 1.19% K. CAFSA with hog manure had a pH of 6.01, with 2.29% OM, 0.70% N, 0.75% P, and 1.05% K. CAFSA with chicken dung had the highest values for pH of 6.40, 3.11% OM, 1.06% N, 1.03% P, and 1.38% K. The results indicated that CAFSA with chicken dung had the highest pH, OM, N, P, and K contents compared to CCAFSA with sow manure and hog manure.

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(13) Conclusions and Recommendations (This part shall be written in narrative form and arranged based on the objectives and findings. Conclusions are general statements answering the objectives. Recommendations are based on the conclusions. They may be split or combined based on their length, which means, conclusions and recommendations may be split under separate headings if they are long. Justify paragraphs with main headings in caps.)

Conclusions

Based on the findings of the study, the following conclusions have been reached:

1. The highest temperature was recorded in variant 2 of CAFSA with chicken dung at 59.66°C on day 16, while the lowest temperature was in variant 2 and variant 2 of CAFSA with hog manure) at 26°C on day 31.
2. All the CAFSA formulations has an Initial moisture content of 8, and started to decline on days 4-5. Variant 1 of CAFSA with hog manure (CAFSA V1) showed a significant drop to 4.44 on day 29, while CAFSA with chicken dung demonstrated the most consistent moisture content in all variants.
3. All CAFSA formulations had an initial increase in pH. Variant 2 of CAFSA with chicken dung exhibited the highly variable pH reaching up to 6.78, while CAFSA with hog manure had the lowest initial pH which does not varies across variants.
4. CAFSA with hog manure had the highest pH and organic matter (OM) in Variant 2, the highest phosphorus in Variant 1, and the highest potassium in Variant 2. CAFSA with cow manure had the highest pH and potassium in V2. CAFSA with chicken dung had the highest pH in V2, the highest OM and phosphorus in V3, and the highest potassium in V2.

Recommendations

Based on the results of this study, the following recommendations are forwarded:

1. Utilize CAFSA with chicken dung, regardless of the variant in growing crops because of its high nutrient content.
2. Conduct long-term studies to understand the impact of different CAFSA variants on soil health and overall fertility.
3. Evaluate the effect of the different CAFSA formulations and variants on the growth and yield of various crops.

(14) Output (Indicate the specific products, processes or services which the project has produced; quantify when possible.)

(15) Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes or services transferred as extension activities)

(16) Economic Analysis (This should present the direct cost of project implementation, expenses in the production of technology, revenues generated from technology produced, return on investment (ROI) and payback period.

(17) Acknowledgments (Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

(18) References (It should include only works mentioned in the text and should be arranged alphabetically. Citations shall not be less than 13 entries. This must follow the APA latest edition format, in hanging indentation with 12-point space in between.)

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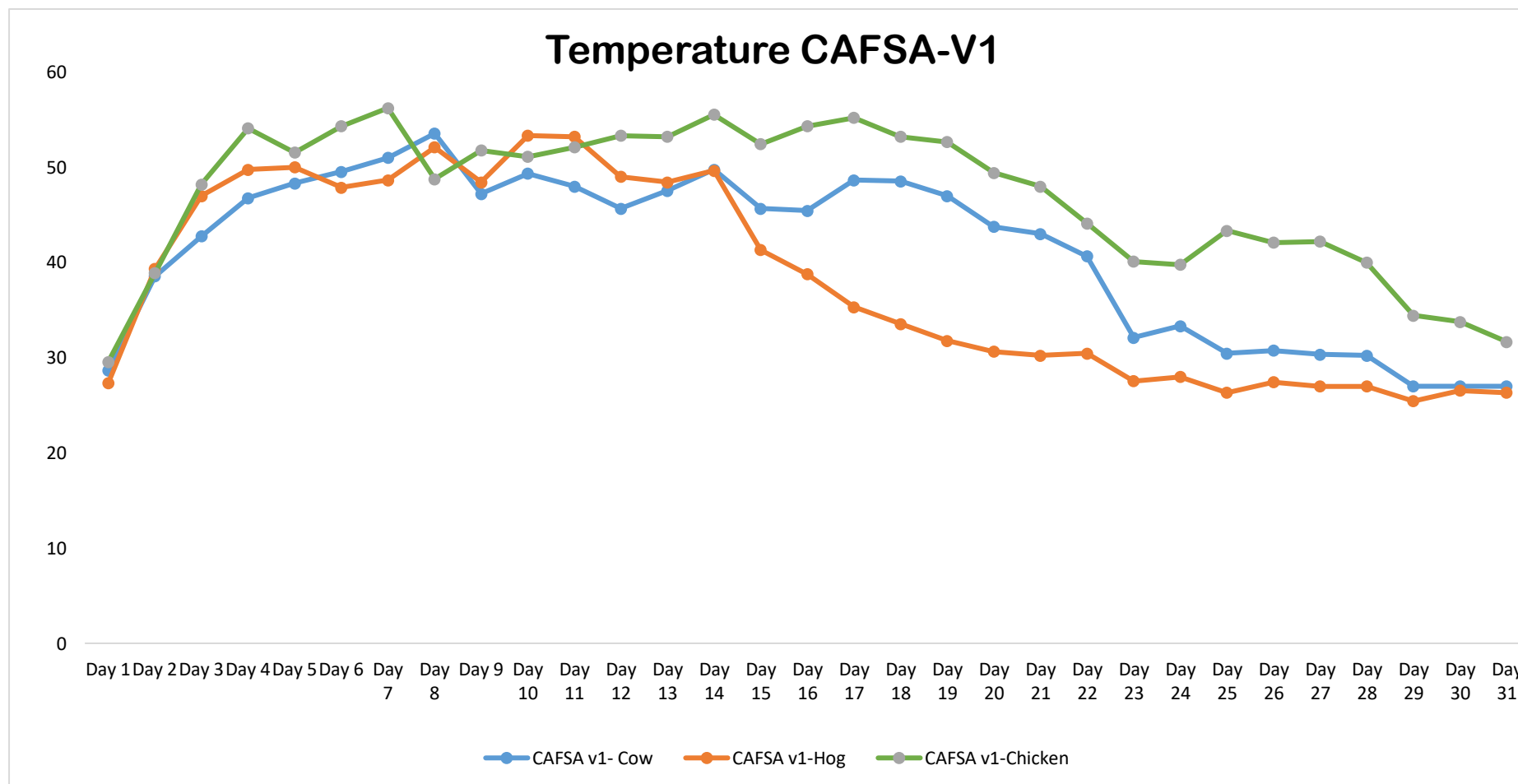


Figure 1. Daily temperature levels of CAFSA V1 for Cow Manure, Hog Manure, and Chicken Dung over 31 days.

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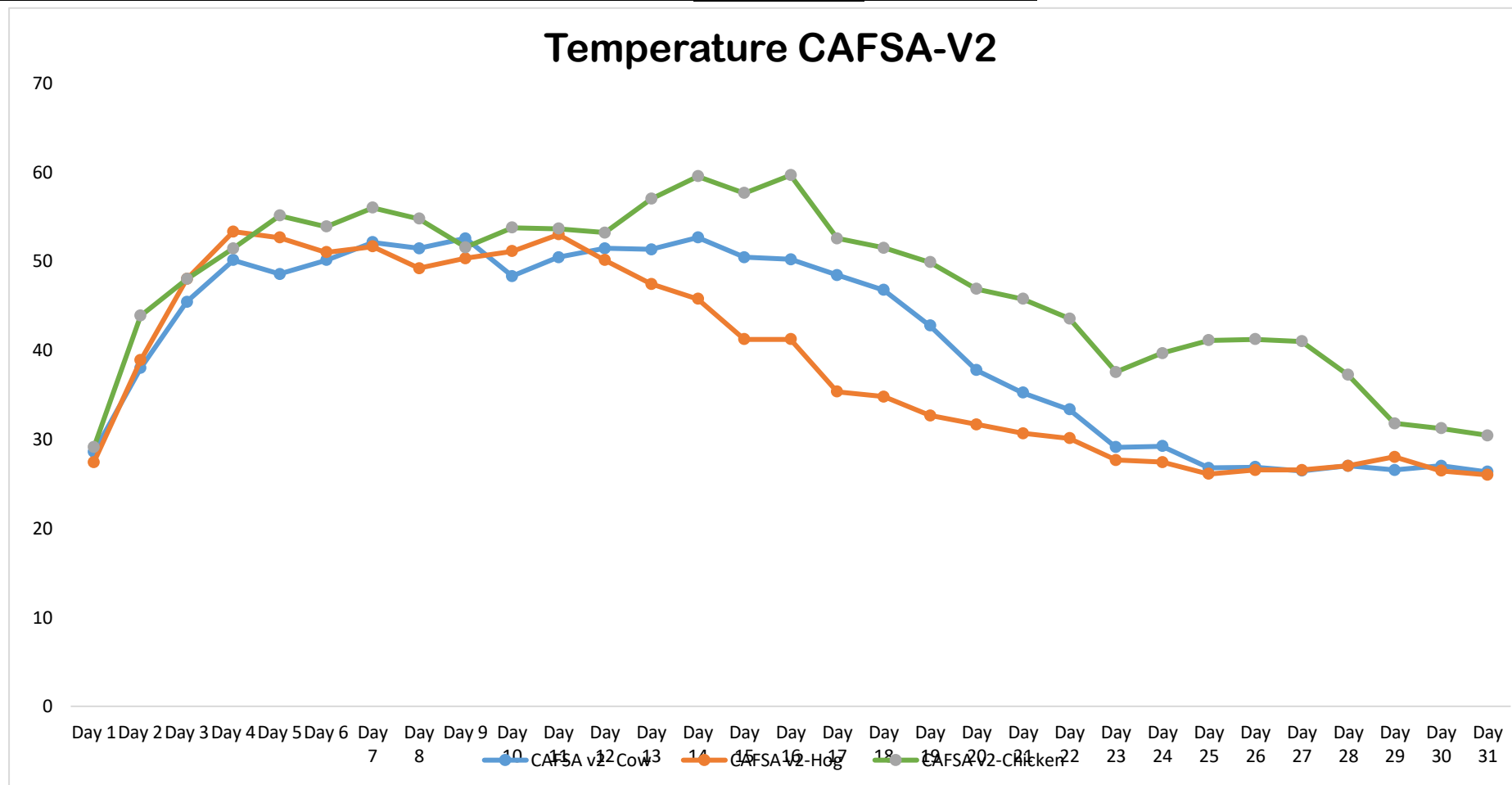


Figure 2. Daily temperature levels of CAFSA V2 for Cow Manure, Hog Manure, and Chicken Dung over 31 days.

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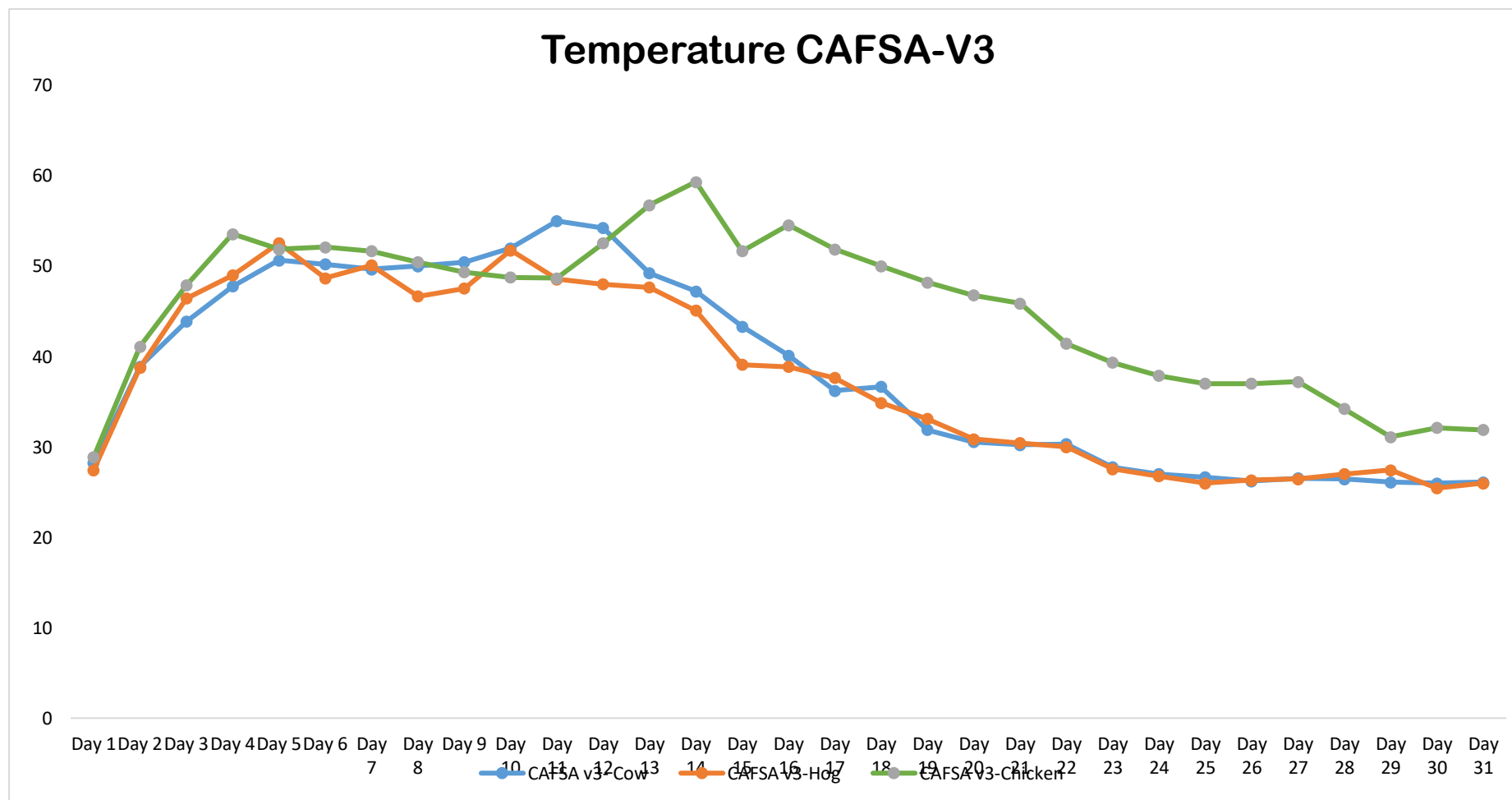


Figure 3. Daily temperature levels of CAFSA V3 for Cow Manure, Hog Manure, and Chicken Dung over 31 days

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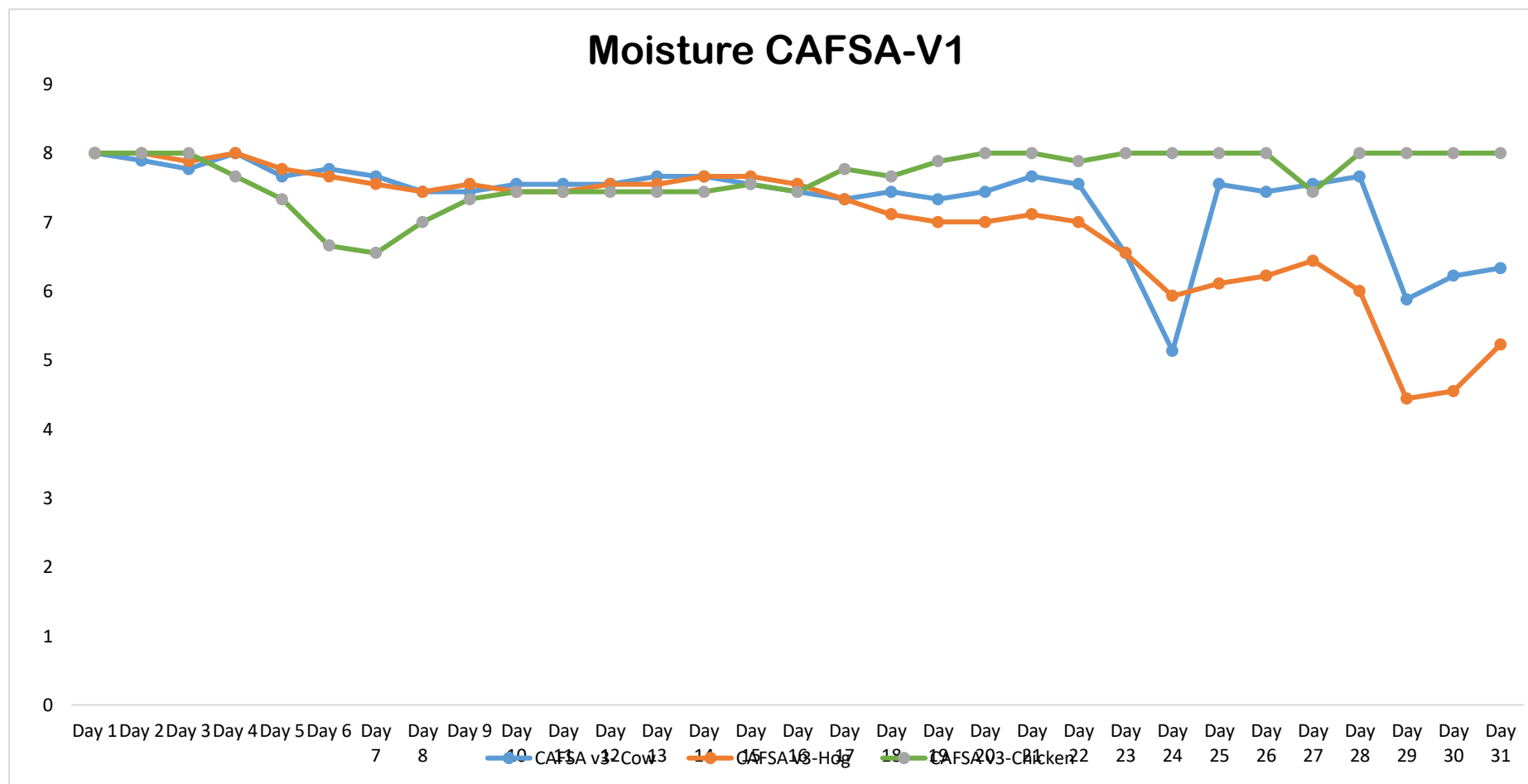


Figure 4. Daily moisture levels of CAFSA V1 for Cow Manure, Hog Manure, and Chicken Dung over 31 days

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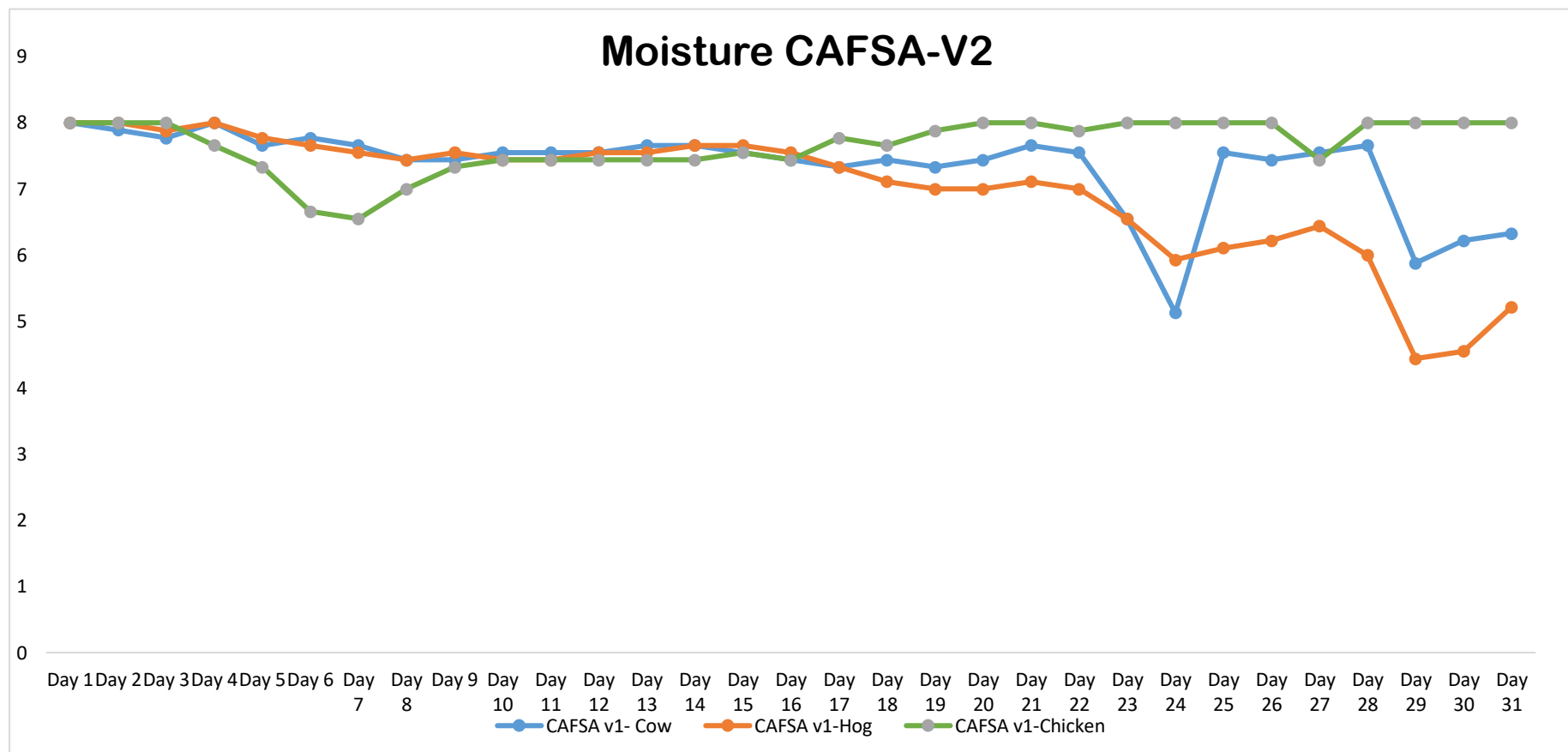


Figure 5. Daily moisture levels of CAFSA V2 for Cow Manure, Hog Manure, and Chicken Dung over 31 days

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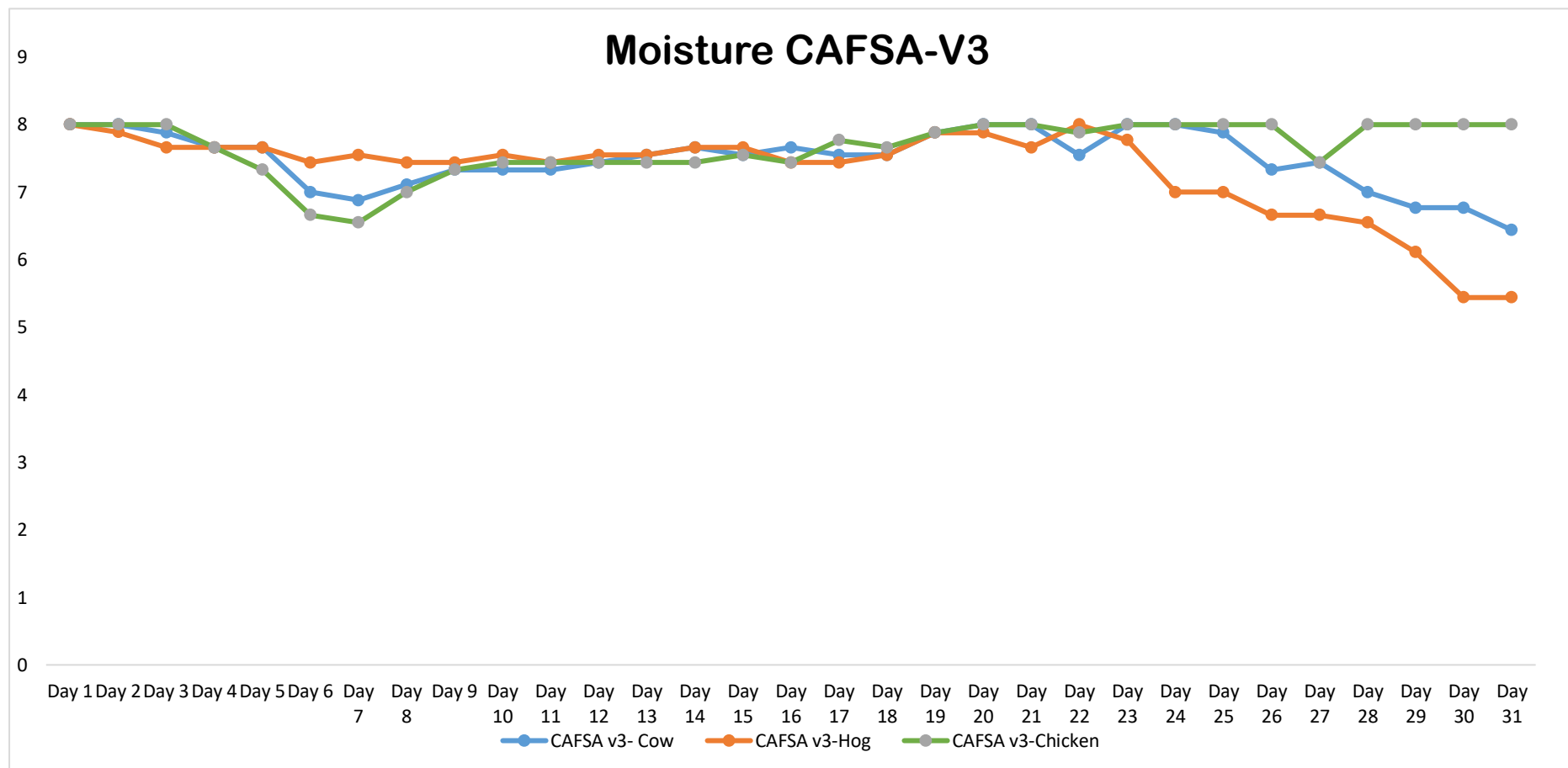
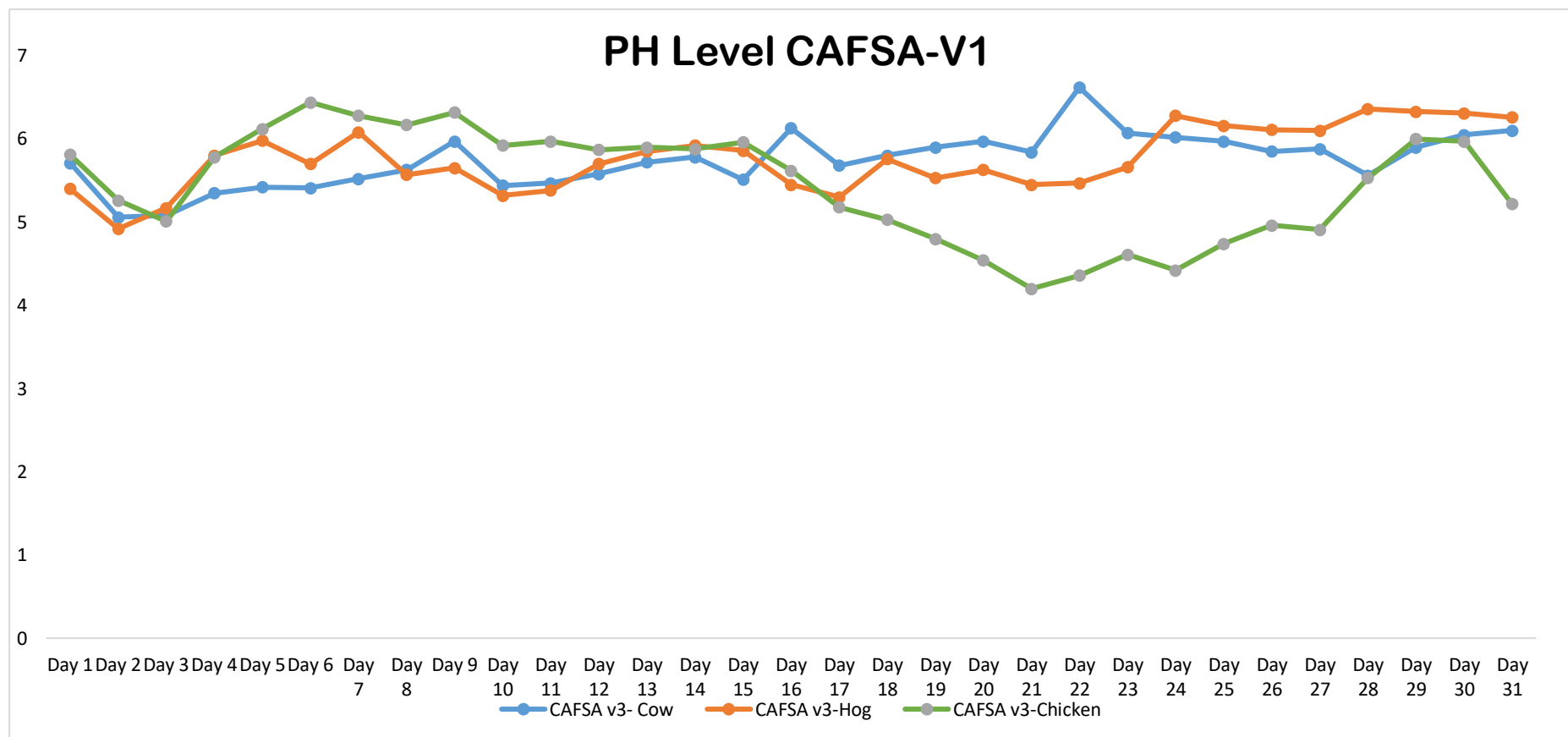


Figure 6. Daily moisture levels of CAFSA V3 for Cow Manure, Hog Manure, and Chicken Dung over 31 day.

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. **Figure 7.** Daily pH levels of CAFSA V1 for Cow Manure, Hog Manure, and Chicken Dung over 31 day

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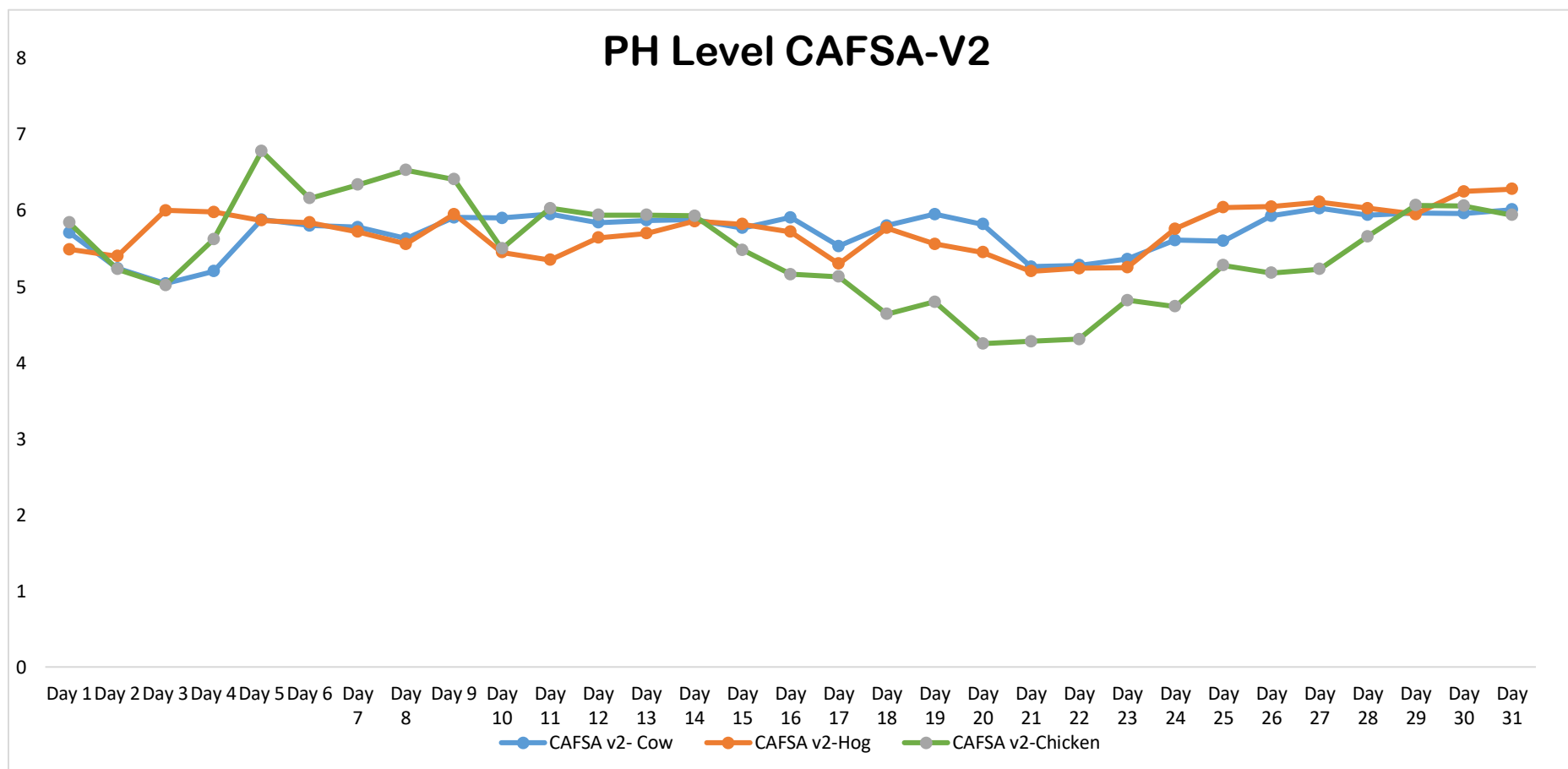


Figure 8. Daily pH levels of CAFSA V2 for Cow Manure, Hog Manure, and Chicken Dung over 31 day.

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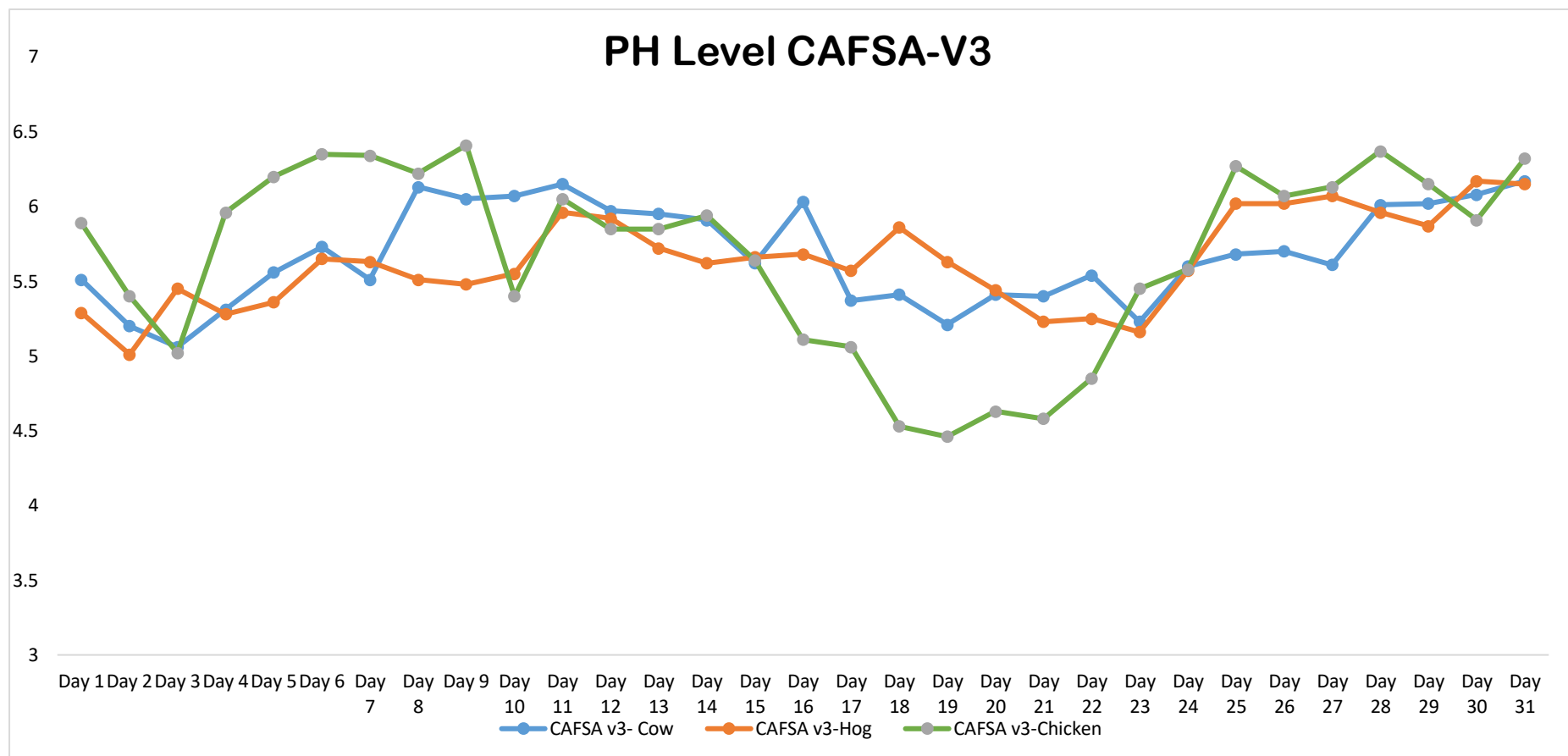


Figure 9. Daily pH levels of CAFSA V3 for Cow Manure, Hog Manure, and Chicken Dung over 31 day.

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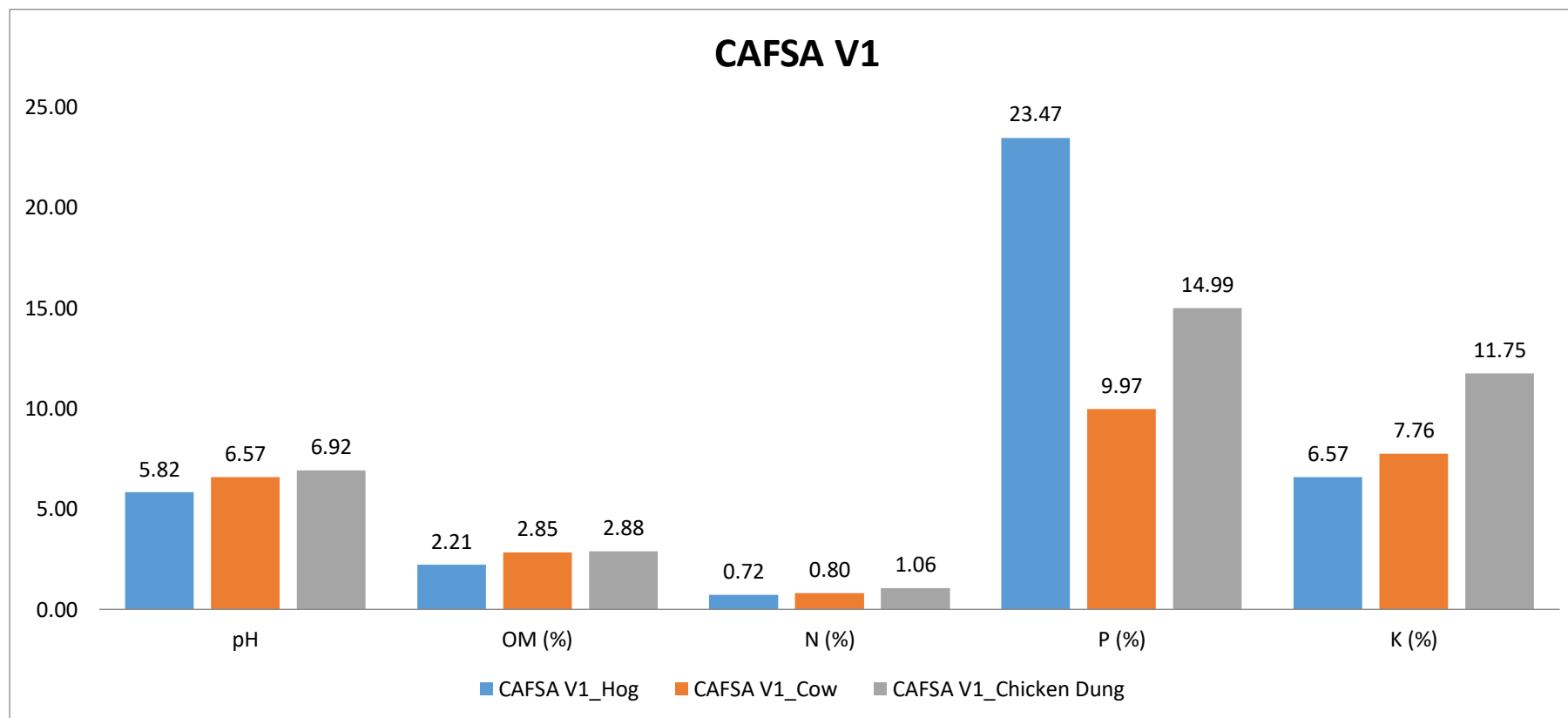


Figure 10. PH, OM, NPK content of CAFSA V1 for Cow Manure, Hog Manure, and Chicken Dung.

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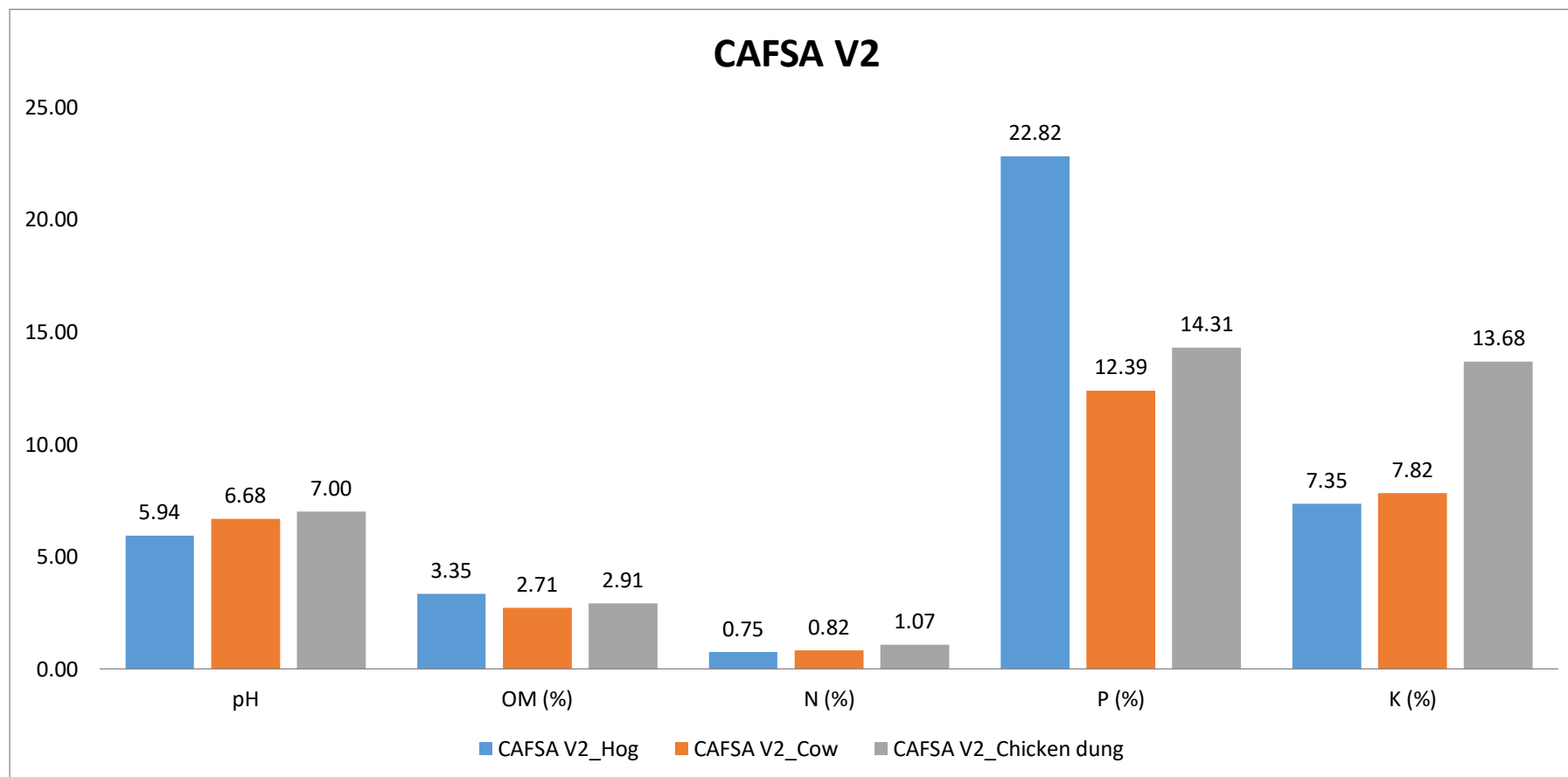


Figure 11. PH, OM, NPK content of CAFSA V2 for Cow Manure, Hog Manure, and Chicken Dung.

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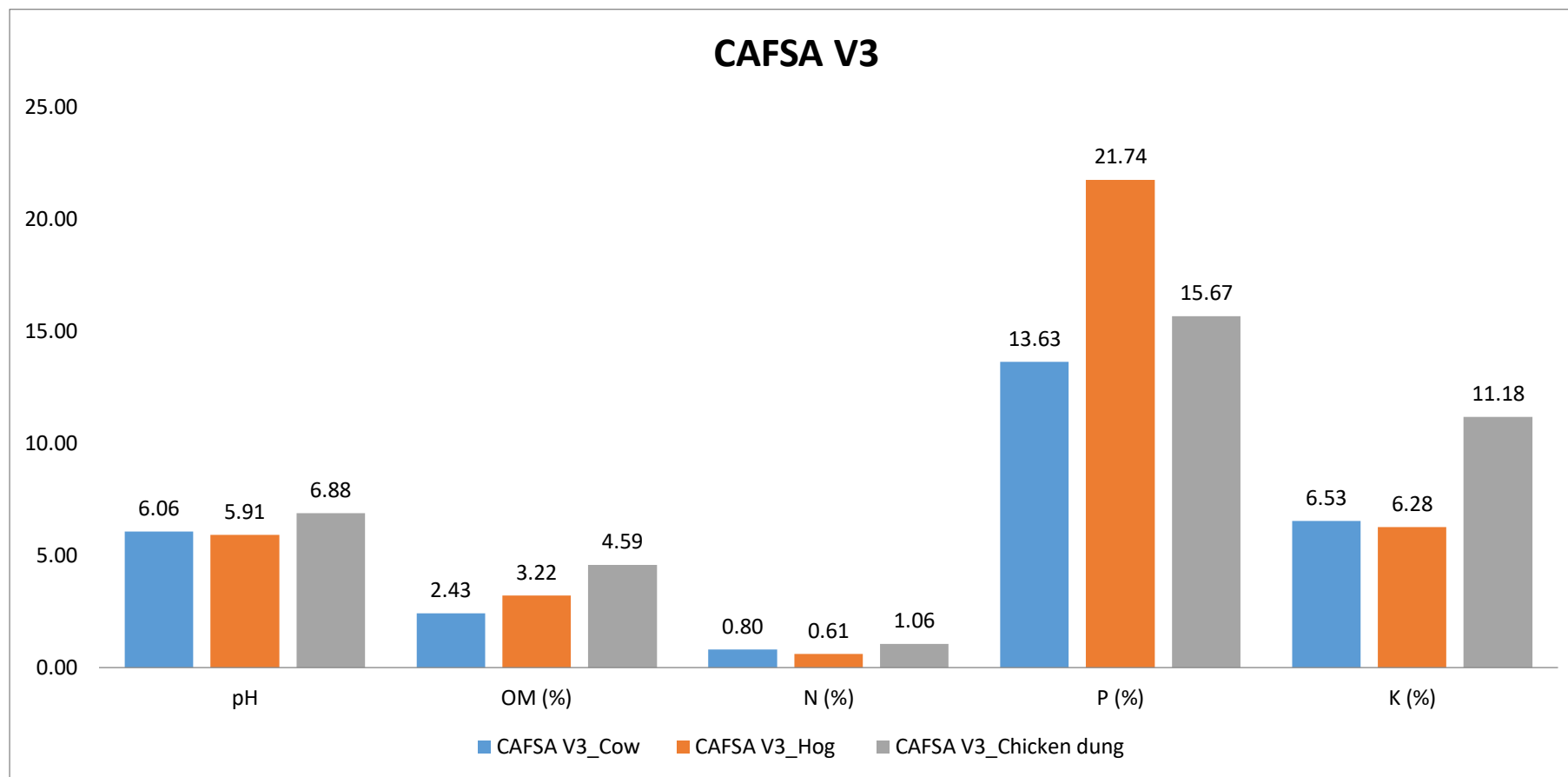


Figure 12. PH, OM, NPK content of CAFSA V3 for Cow Manure, Hog Manure, and Chicken Dung.

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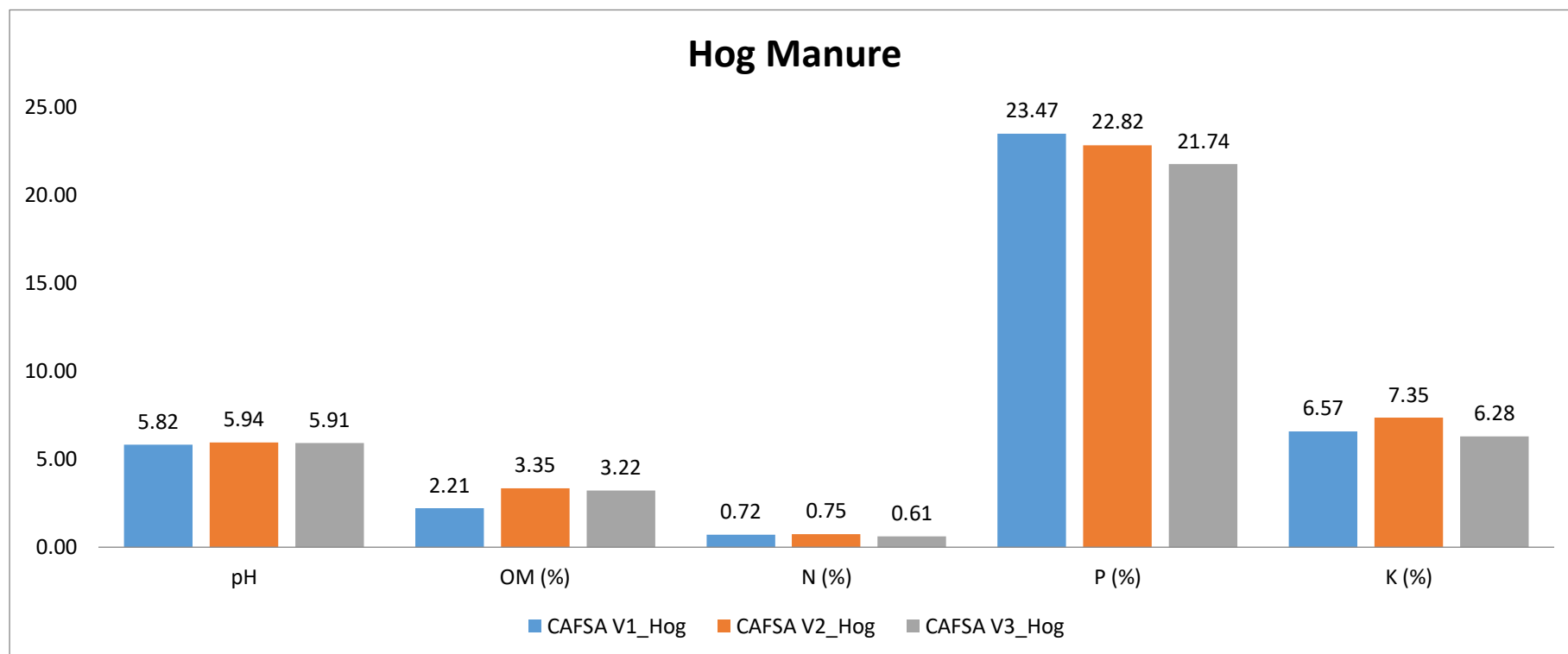


Figure 13. PH, OM, NPK content of Hog manure in CAFSA V1, CAFSA V2 and CAFSA V3.

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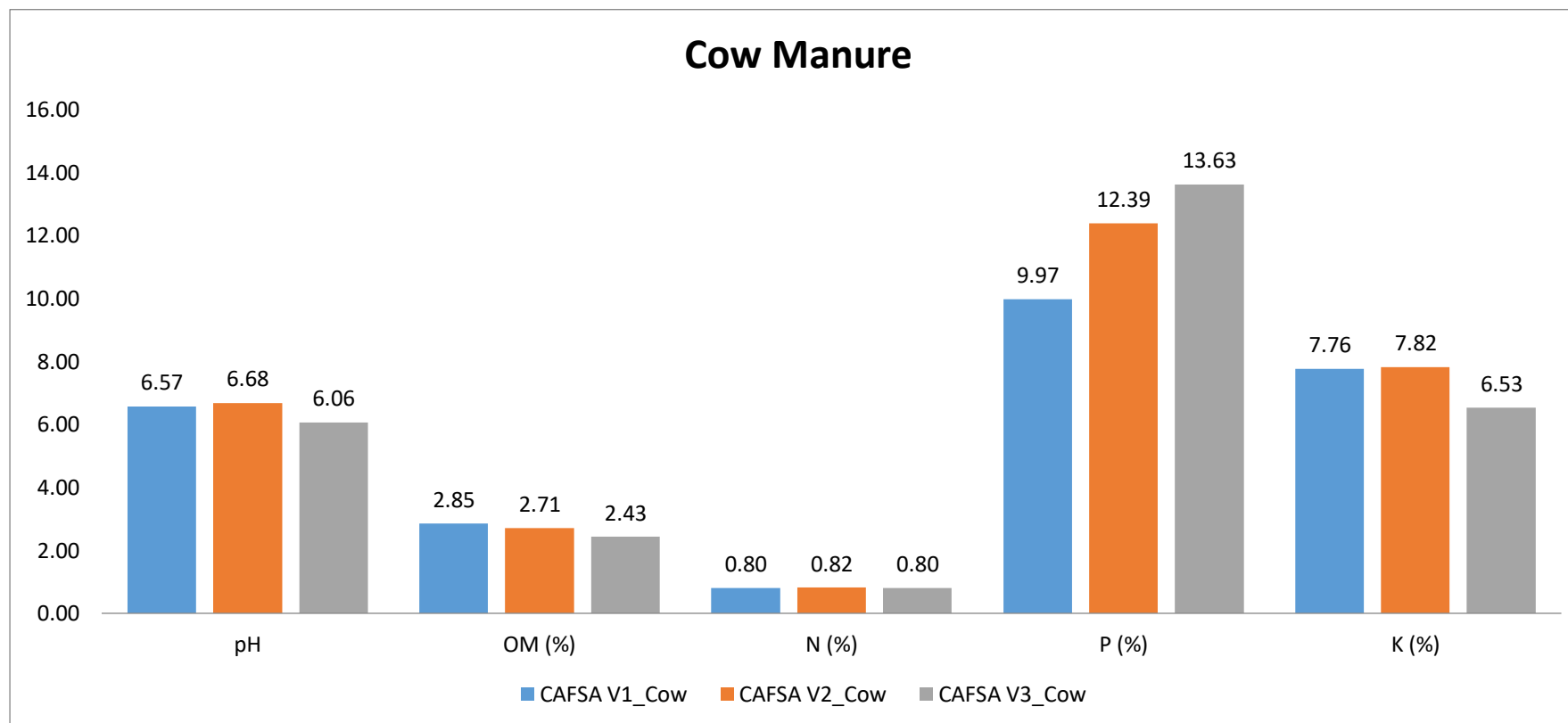


Figure 14. PH, OM, NPK content of Cow manure in CAFSA V1, CAFSA V2 and CAFSA V3.

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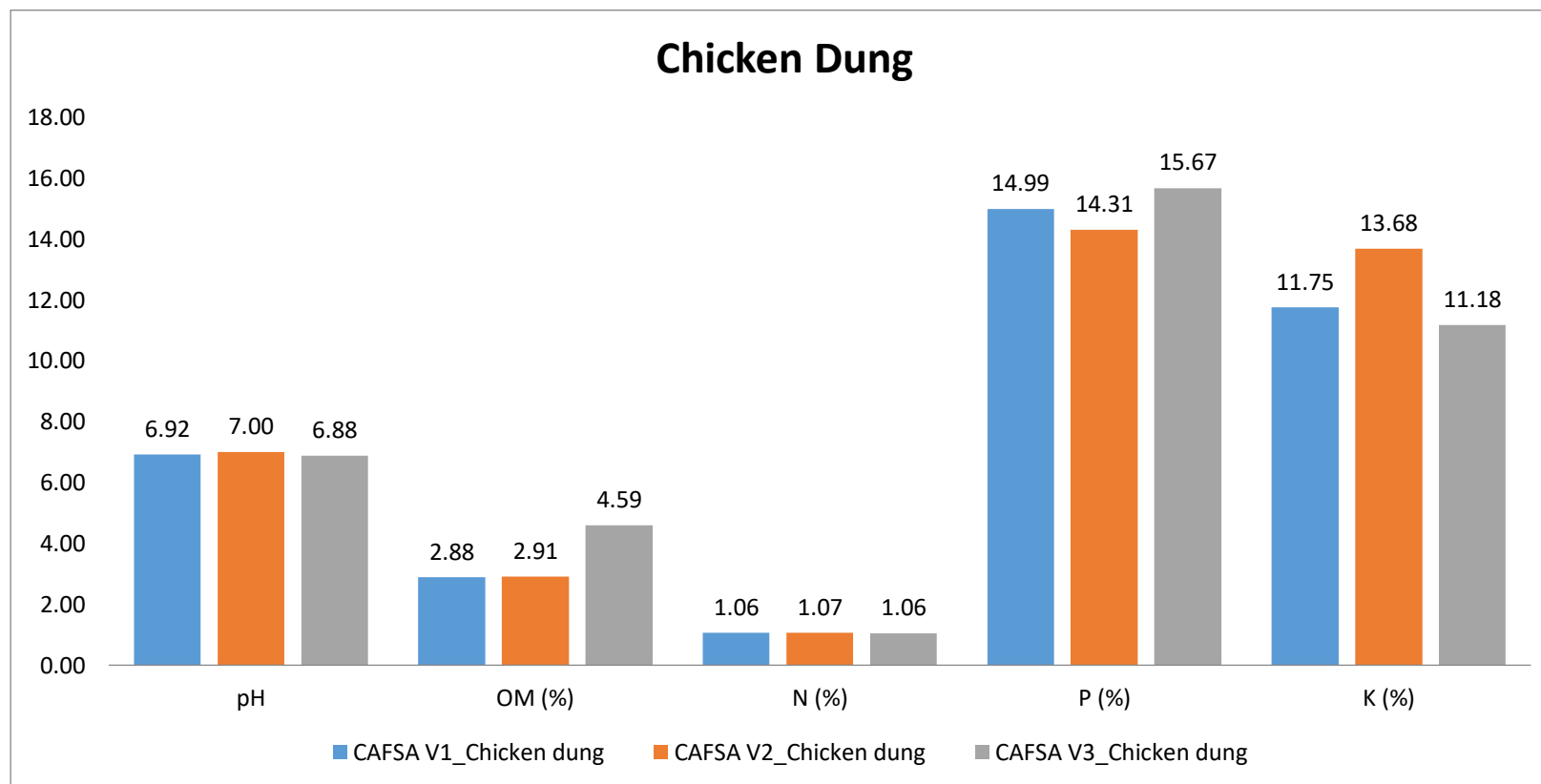


Figure 15. PH, OM, NPK content of Chicken dung in CAFSA V1, CAFSA V2 and CAFSA V3.